CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-503: Computer Networks

Session: December 2023 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Explain the use of Bridges in Computer Networks.

Definition: A **Bridge** is a Data Link Layer (Layer 2) device that connects two separate LAN segments running the same protocol, inspecting MAC addresses to filter frames and segment collision domains.

Q2 (3M) Write a short note on E-Mail and its core protocols.

Explanation: Electronic mail architecture consists of User Agents (UAs) and Message Transfer Agents (MTAs). Uses SMTP for sending emails, and POP3 / IMAP for downloading emails to client devices.

Q3 (3M) What is Network Topology? Name major topologies.

Topologies: The geometric layout of interconnected nodes: Star, Bus, Ring, Mesh, Tree, and Hybrid.

Q4 (3M) What is Attenuation in transmission media?

Definition: The progressive loss of signal energy/strength as it propagates through transmission media, measured in decibels (dB), compensated using amplifiers or repeaters.

Q5 (3M) Define Bandwidth and Throughput.

Definitions:

- **Bandwidth:** The maximum theoretical data-carrying capacity of a communication channel (bits per second).
- **Throughput:** The actual rate of successfully transmitted data delivered across the network per unit time.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) Describe the functions & responsibilities of the Session Layer.

Session Layer (Layer 5):

- **Session Establishment, Maintenance, and Termination:** Manages dialogs between communicating processes.
- **Dialog Control:** Determines whether communication is simplex, half-duplex, or full-duplex.
- **Synchronization / Checkpointing:** Inserts sync tokens into data streams to allow recovery from checkpoints in case of interruptions.

Q7 (7.5M) **Differentiate between OSI Model and TCP/IP Model.**

Comparison:

Aspect	OSI Reference Model	TCP/IP Model
Layers	7 layers	4 layers
Origin	Theoretical standard created by ISO.	Practical protocol suite developed for ARPANET.
Session & Presentation	Separate distinct layers.	Integrated into Application layer.
Protocols	Protocols developed after model.	Model developed based on existing protocols.

Q8 (7.5M) **Explain IP Addressing and Subnetting.**

IP Addressing: Classes A, B, C, D, E. Subnetting borrows host bits to divide large networks into smaller subnets.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **What is Congestion Control? How is Congestion Control different from Flow Control? Explain various congestion control techniques.**

Congestion vs Flow Control:

- **Flow Control:** Point-to-point mechanism preventing a fast sender from overwhelming a single slow receiver (Sliding Window).
- **Congestion Control:** Network-wide mechanism preventing global traffic load from exceeding overall subnet router buffer capacity.

Techniques: Leaky Bucket algorithm, Token Bucket algorithm, Choke Packets, and Random Early Detection (RED).

Q10 (15M) **Explain Error Control in Data Link Layer using CRC with a complete numerical example.**

CRC Numerical Example:
Data: `1010000`, Generator: `1001`.
Append 3 zeros: `1010000000`. Perform modulo-2 binary division by `1001` to calculate remainder (FCS). Complete step-by-step arithmetic shown.

Q11 (15M) **Explain Dijkstra's Shortest Path Routing Algorithm with a network graph example.**

Dijkstra's Algorithm: Greedy shortest path calculation from source router to all other nodes maintaining visited sets and updating tentative distances.

Q12 (15M) **Explain Transmission Media: Twisted Pair, Coaxial Cable, and Optical Fiber.**

Guided Media: Bandwidth, attenuation, electromagnetic immunity, and cost trade-offs across all three physical media.

Q13 (15M) **Write short notes on: (a) Firewalls, (b) FTP, (c) Domain Name System (DNS).**

Detailed Notes: Packet filtering firewalls, dual-port FTP architecture (Control port 21, Data port 20), and hierarchical DNS domain resolution.
